

CODEALPHA MACHINE LEARNING INTERNSHIP

CREDIT SCORING MODEL USING MACHINE LEARNING

Submitted By:

Shaik Nazma

Internship Domain:

Machine Learning

Organization:

CodeAlpha

Submission Date:

June 2026

CREDIT SCORING MODEL USING MACHINE LEARNING

1. Introduction

Credit scoring is an important application of Machine Learning in the banking and financial sectors. Financial institutions use credit scoring systems to evaluate the creditworthiness of loan applicants before approving loans. By analyzing historical financial information and applicant details, machine learning models can assist in making accurate and data-driven lending decisions. This project implements a Credit Scoring Model using machine learning algorithms to predict loan approval status based on applicant information.

2. Objective

The main objective of this project is to develop a machine learning model capable of predicting an individual's creditworthiness using historical financial data. The project also compares the performance of different classification algorithms and evaluates their effectiveness using multiple performance metrics.

3. Dataset Description

The project uses the Loan Prediction Dataset, which contains applicant demographic and financial information.

Dataset Details

- Total Records: 614
- Total Features: 11 predictive attributes
- Target Variable: Loan Status (Approved / Not Approved)

Features Used

- Gender
- Married Status
- Dependents
- Education
- Self Employed
- Applicant Income
- Coapplicant Income
- Loan Amount
- Loan Amount Term

- Credit History
- Property Area

The dataset contains both categorical and numerical attributes commonly used in loan approval decision-making.

4. Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

5. Methodology

Step 1: Data Collection

The Loan Prediction Dataset was loaded into Python using the Pandas library.

Step 2: Data Preprocessing

Several preprocessing steps were performed:

- Missing value handling
- Data cleaning
- Categorical data encoding using Label Encoder
- Feature scaling using StandardScaler

Step 3: Train-Test Split

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

Step 4: Model Training

Two machine learning classification algorithms were implemented:

1. Logistic Regression
2. Random Forest Classifier

Step 5: Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score
- Confusion Matrix

6. Results

Logistic Regression

- Accuracy: 78.86%

Random Forest Classifier

- Accuracy: 74.80%

ROC-AUC Score

- 0.672

Classification Report Summary

Metric	Value
Precision	83%
Recall	79%
F1-Score	76%

Confusion Matrix

```
[[18 25]
```

```
[ 1 79]]
```

The Logistic Regression model achieved the highest prediction accuracy and outperformed the Random Forest model on this dataset.

7. Visualizations

The following visualizations were generated:

- Confusion Matrix Heatmap
- Model Accuracy Comparison Graph

These visualizations helped analyze model performance and compare classification accuracy.

8. Conclusion

This project successfully developed a Credit Scoring Model using Machine Learning techniques. The dataset was preprocessed, encoded, and used to train Logistic Regression and Random Forest models. Logistic Regression achieved the highest accuracy of 78.86%, demonstrating its effectiveness in predicting loan approval status. The project highlights the practical application of machine learning in credit risk assessment and financial decision-making.

9. Future Enhancements

- Use larger and more diverse financial datasets.
- Apply advanced ensemble learning techniques.
- Implement XGBoost and Gradient Boosting models.
- Develop a web-based loan approval prediction system.
- Integrate real-time financial and credit history data.